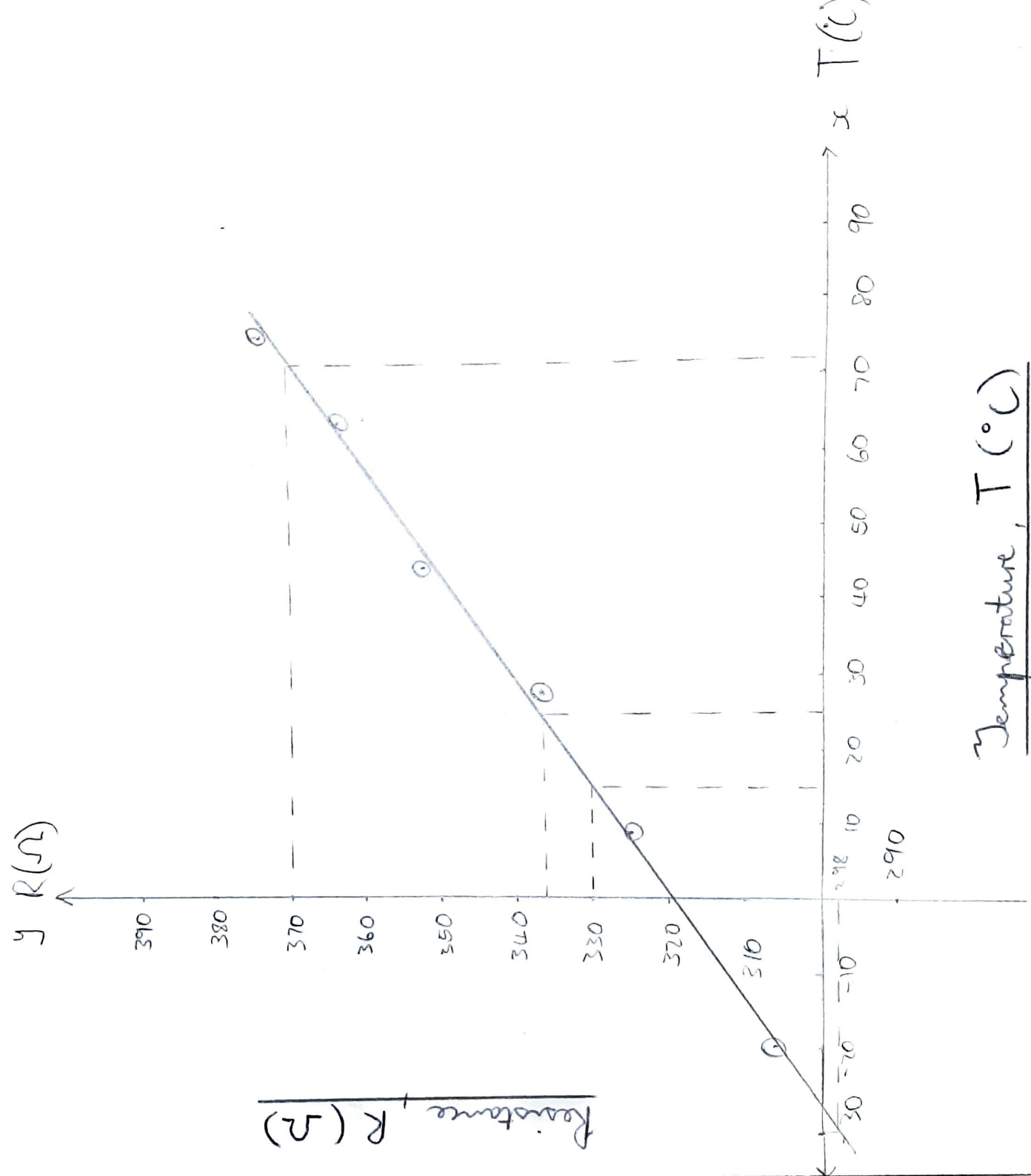


The resistance R_x of a wire at various temperatures, T



Now that you have drawn a graph of the data and found that there is a linear relationship between the variables, you can formulate the mathematical relationship between the variables. This mathematical formula is called the empirical relationship since it was determined using experimental data.

2. Determine and clearly indicate on your graph, the change in resistance per unit change in the temperature, that is $\Delta R/\Delta T$, or the slope of the graph. Choose points 1 and 2 as far as possible from each other - see figure 1 - and **do not** use data points from the table. Do not forget the units of the gradient when you state its value.

$\frac{\Delta R}{\Delta T} = \frac{370 - 330}{71 - 15}$	Point 1 (330 ; 15; 330)
$= \frac{5}{7} \text{ ohm}/^{\circ}\text{C}$	Point 2 (71; 370)

3. Set up the empirical equation for the resistance of the wire at different temperatures, i.e., the formula relating the relationship between the resistance and temperature of the wire. This equation should have values and units for both constants in the linear equation. Use the symbols R and T for the variables in the equation.

$R = \frac{5}{7}T + c$	$\therefore R = \frac{5}{7}T + 319,29$
$330 = \frac{5}{7}(15) + c$	$\rightarrow$
$c = 319,29$	Where $T = \text{Temperature } (^{\circ}\text{C})$
	$R = \text{Resistance } (\Omega)$

4. Now that you have determined the relationship between the measured quantities, you can either use the graph or the empirical equation to determine values for the resistance of the wire at other temperatures. Let's use the graph to determine the resistance of the wire at some temperature. You should indicate **on the graph** where the readings were taken:

- The resistance of the wire at -30°C . You are extrapolating in this case. That is, you are assuming that the functional relationship between the variables will remain linear outside of the measured range. Such an assumption is risky since you do not know whether the relationship is still linear outside of the measured range of values.
- The resistance of the wire at 25°C . Now you are interpolating, or predicting that the resistance will take on this value, a much safer bet since this is within the measured range of values.

i) 298Ω

ii) 336Ω

2. The power function: $y = kx^n$

The power function is often found in physics, for example:

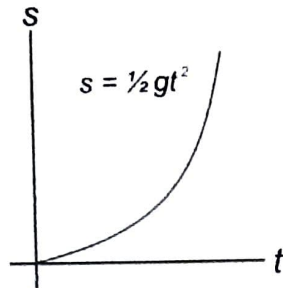


Figure 3: Freefall out of

- a. The distance, s , that an object, initially at rest, moves under the influence of gravity during a time interval t is given by:

$$s = \frac{1}{2} gt^2$$

- b. The period, T , of a simple pendulum depends on its length, L , according to:

$$T = k\sqrt{L}$$

- c. The force, F , between two point charges or masses separated by a distance, r , is given by an inverse square law:

$$F = k \frac{1}{r^2}$$

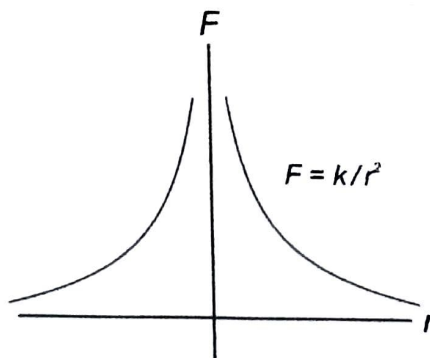


Figure 4: The electric force between two point charges

Advantages of linear representations

- a. It is easy to fit data points to a straight line.

Gradient = n

$$= \frac{\log(a_2) - \log(a_1)}{\log(T_2) - \log(T_1)}$$

$$= \frac{\log(a_2/a_1)}{\log(T_2/T_1)}$$

$$= \frac{\log(12952) - \log(527)}{\log(550) - \log(4.57)}$$

$$= \frac{\log(12952/527)}{\log(550/4.57)}$$

$$= 0.6668$$

$$\therefore n \approx 0.67$$

3. Read the y-intercept of the graph off where the period has the value $T = 1.0$ day (why?). This is the constant k in the equation above. Why don't we take the log in this case? The units of k can be determined by considering the units of the variables, a and T , assuming the relationship given in equation (2).

$$K = 180 \text{ Where } T = 1$$

4. Next, set up the empirical relationship between the distance and period values. Supply values for the constants, together with their units, in the equation.

$$a = kT^n$$

$$\therefore a = 180T^{0.67}$$

$$T = \text{time (days)} \quad a = \text{distance} (\times 10^3 \text{ km})$$

5. What can you conclude about the mathematical relationship between a and T , according to your graphical analysis?

a and T are directly proportional to each other.

The semi-major axis a and period T of orbit of some of Saturn's moons.

